# 1.Introduction:

## a)Background And Context:

Desktop voice assistants,like Siri, Google Assistant and many voice assistants have evolved from early speech recognition systems in 1950’s and 1960’s.These assistants use voice recognition,language processing algorithms and voice synthesis to understand and respond to user commands.These voice assistants have become integral in personal and business settings transforming human-computer interactions and voice user interfaces.

## b)Problem statement and Objectives:

The project aims to address challenges in daily tasks,such as improving efficiency,multitasking and accessibility.They streamline time,enhance productivity,and offer hands-free operation.The objectives of the project are taking tasks like managing schedules,setting reminders,answering queries and controlling smart devices.

## C)significance and Motivation:

Voice assistants hold significant importance in daily life by enhancing efficiency, accessibility,and user experience across various industries like healthcare,retail,and finance.Additionally , voice assistants are evolving with advancements in natural language processing,and expanding their role in our daily lives.

# 2.Project Overview :

## a)Project Goals and Scope:

• Develop a desktop voice assistant for managing several tasks in a system environment.

•Improving productivity and hands-free operation to interact with daily tasks.

## b.Project Timeline:

**Week-1: Finding out requirements and probability of working on the project.**
**Week-2: Making the framework for the project using [tkinter and various functions].**
**Week-3: Importing speech and text recognition modules ,making the functions work.**
**Week-4: Implementing action buttons for various commands and testing.**

## c.Project repository:

## Project link

## d.Team Members and Contributions:

### 1)M.Ajay Kumar:

**Created the GUI(graphical unified interface) from the tkinter and build the structures of the buttons and implemented the button functions of the framework Imported all the framework needed modules and defined the image for the voice assistant.**

### 2)K.Hemanth Charan Ganesh:

**Gave the various commands for the action buttons in the interface and system required commands for the interface to deal with the tasks need to overlay the Project objectives.**

### 3)V.Nandeeshwar:

**Imported the voice recognition and text recognition modules from the python websites and defined their functions to the user data to work effectively with the microphone for the project objectives,and gave the required commands for functioning.**

# 3.Methodology:

## a.Approach and Methodology Employed:

**The project follows an initiative development of the voice assistant and gradually implementing various commands for the user interface and hands-free control of the tasks .**

## b.Tools, Technologies,and Modules Used:

**.tkinter module for the framework and designing of the action buttons.**

**.pyaudio module for the voice and text recognition.**

# 4.Results and Analysis:

## a)Achieved Results and Findings:

**.Developed a functional voice assistant for desktop**

**.Designed a user-friendly interface for task management in system**

## b) Discussion of Challenges and Solutions

**Challenge:**

**We have faced a lot of issues while converting text to speech.Initially we have used gTTS(google text to speech) but onl;y works if it has only internet connection.**

**Solution:**

**Instead of gTTS we have used pyttsx3 to convert text to speech.This also works in offline too.**

## 5) Conclusion and Future development:

a) **Summary of Outcomes and Contributions:**

**The project has successfully delivered a functional voice assistant leading specified requirements .Users can conveniently manage tasks hands-free through the interface provided.**

b) **Assessment of project Success:**

**The project seems successful based on achieving its objectives within defined functions.User feedback and adoption rates will further validates its success.**

c) **Recommendations for future development:**

**Implementing additional voice commands for the industrial and management usage.**

d) **Project Team Members Git Accounts:**

**1)Hemanth**

**2)Ajay**

**3)Nandeeshwar**

## 6)References:

https://pypi.org/project/PyAudio/

https://docs.python.org/3/library/tkinter.html#module-tkinter

https://pypi.org/project/pyttsx3/